

Marathwada Mitra Mandal's College of Engineering
Karvenagar, Pune- 52

An Autonomous Institute



Curriculum for
Second Year B. Tech Information Technology
A.Y. 2025-26

Permanently Affiliated to SPPU | Accredited with A++ grade by NAAC
Recipient of Best College award by SPPU | Accredited by NBA
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Institute Vision

To aspire for the Welfare of Society through excellence in Science and Technology.

Institute Mission

Our Mission is to

- Mould young talent for higher endeavors.
- Meet the challenges of globalization.
- Commit for social progress with values and ethics.
- Orient faculty and students for research and development.
- Emphasize excellence in all disciplines

Department Vision

To emerge as a center of excellence in education, research and innovation in Information Technology for enrichment of society.

Department Mission

- To cater industry with engineers having theoretical & practical background and competent IT skills.
- To pursue advanced knowledge in the field of information technology.
- To inculcate budding IT engineers with professional and interpersonal skills.

Program Educational Objectives (PEOs)

1. Adequate knowledge and skills in Information Technology for implementation of complex problems with innovative approaches.
2. Inclination and technical competency towards professional growth in the field of Information Technology.
3. Ethics and value based interpersonal skills to facilitate lifelong learning and societal contributions.

Program Outcomes (POs)

Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization for the solution of complex engineering problems.

Problem analysis: Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety, and cultural, societal, and environmental considerations.

Conduct investigations of complex problems: The problems that cannot be solved by straightforward application of knowledge, theories and techniques applicable to the engineering discipline.

Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modeling to complex engineering activities, with an understanding of the limitations.

The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

Communication: Communicate effectively on complex engineering activities with the engineering community and with the society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change

Program Specific Outcomes (PSOs)

A graduate of the Information Technology Program will be able to

1. Develop quality computer applications by applying principles of software engineering.
2. Pursue advancement in the field of data engineering.

Abbreviations

BSC:	Basic Science Course
UG:	Undergraduate Programme
ESC:	Engineering Science Course
PCC:	Program Core Courses
PEC:	Program Elective Courses
MDM:	Multidisciplinary Minor Courses
OE:	Open Elective
VSC:	Vocational Skill Course
SEC:	Skill Enhancement Course
VEC:	Value Educational Course
AEC:	Ability Enhancement Course
HSSM:	Humanity and Social Science Management
IKS:	Indian Knowledge System
VEC:	Value Education Course
FP:	Field Project
CC:	Co-Curricular Course
IT:	Internal Tool
ET:	External Tool
ETE:	End Term Examination
CIE:	Continuous Internal Evaluation
TW:	Term work
OR:	Oral Examination
PR:	Practical Examination
L:	Lecture
P:	Practical
T:	Tutorial
OL:	Online Teaching
ODL:	Open Distance Learning

Second Year B.Tech Information Technology Semester-III															
Course Code	Course Name	Course Type	Teaching Scheme(Hrs./week)			Examination Scheme						Credits			
			L	P	T	CIE	ETE	TW	PR	OR	Total	L	P	T	Total
B25IT301	Data Structures and Algorithms	PCC	3	2	-	40	60	-	50	-	150	3	1	-	4
B25IT302	Object Oriented Programming	PCC	3	2	-	40	60	-	50	-	150	3	1	-	4
B25IT303	Design Thinking for User Experience: UI/UX	PCC	3	-	-	40	60	-	-	-	100	3	-	-	3
B25IT304	Experiential Learning Courses: Project Based Learning	FP	-	4	-	-	-	50	-	-	50	-	2	-	2
B25IT305	Logic Design & Computer Organization	MDM	2	2	-	40	60	25	-	-	125	2	1	-	3
B25IT306	Problem Solving and Analytical Skills	HSSM-VEC	-	-	2	-	-	50	-	-	50	-	-	2	2
B25IT307	Professional Communication and Analytical Skills	AEC	-	-	2	-	-	50	-	-	50	-	-	2	2
Total			11	10	4	160	240	175	100	-	675	11	5	4	20

Second Year B.Tech Information Technology Semester-IV															
Course Code	Course Name	Course Type	Teaching Scheme(Hrs./week)			Examination Scheme						Credits			
			L	P	T	CIE	ETE	TW	PR	OR	Total	L	P	T	Total
B25IT401	Database Management Systems	PCC	3	2	-	40	60	-	50	-	150	3	1	-	4
B25IT402	Operating Systems	PCC	3	2	-	40	60	-	50	-	150	3	1	-	4
B25IT403	Computer Network	PCC	3	-	-	40	60	-	-	-	100	3	-	-	3
B25IT404	Foundation of Web Technology	OE	3	-	-	40	60	-	-	-	100	3	-	-	3
B25IT405	Web Development & Hosting	SEC	-	4	-	-	-	50	-	-	50	-	2	-	2
B25IT406	Environmental Sustainability	HSSM-VEC	-	-	2	-	-	50	-	-	50	-	-	2	2
B25IT407	Startup Fundamentals and Financing	HSSM	-	-	2	-	-	50	-	-	50	-	-	2	2
Total			12	8	4	160	240	150	100	-	650	12	4	4	20

Abbreviations:

L- Lecture

P- Practical

T- Tutorial

CIE-Continuous Internal Evaluation

ETE- End Term Examination

TW- Term work

PR- Practical

OR- Oral

L : 1 Hr.= 1 credit

P: 2 Hr. = 1 Credit

T: 1 Hr. = 1 Credit

Second Year B. Tech Information Technology

Spring Semester

Course Code: B25IT301

Course Name: Data Structures and Algorithms

Teaching Scheme (Hours/Week)					Examination Scheme						Credits			
L	P	T	OL	ODL	CIE	ETE	TW	PR	OR	TOTAL	L	P	T	TOTAL
3	2	-	-	-	40	60	25	25	-	150	3	1	-	4

Course Outcomes:

After completing the course, the students will be able to:

CO1: Analyze algorithms with respect to time and space complexity

CO2: Select appropriate searching and/or sorting techniques in the application development

CO3: Apply the concept of Abstract Data Types to represent and manipulate data structures

CO4: Apply learned algorithm and data structures to solve the problems

CO5: Design and implement different hashing functions

Unit	Contents	Hrs.
1	Introduction: Introduction to Data Structures: Concept of data, Data object, Data structure, Concept of Primitive and non-primitive, linear and nonlinear, static and dynamic, persistent and ephemeral data structures, Definition of ADT Analysis of algorithms : Frequency count and its importance in analysis of an algorithm, Time complexity and space complexity, Big 'O', 'Ω' and 'θ' Linked organization: Concept of linked organization, Singly Linked List, Doubly Linked List, Circular Linked List (Operations: Create, Display, Search, Insert, Delete)	8
2	Searching and Sorting Searching and Sorting: Need of searching and sorting, Concept of internal and external sorting, sort stability, Searching methods: Linear and binary search algorithms, Sorting methods: Bubble, insertion, Quick, Merge Hashing: Hash tables and scattered tables: Basic concepts, hash function, characteristics of good hash function, Different key-to-address transformations techniques, synonyms or collisions, collision resolution techniques- linear probing, quadratic probing, rehashing, chaining with and without replacement	8
3	Stack and Queues Stack: Concept of stack, Concept of implicit and explicit stack, stack as an ADT using sequential and linked organization, Applications of stack: recursion, converting expressions from infix to postfix or prefix form, evaluating postfix or prefix form, Queue: Concept of queues as ADT, Implementation of queue using array and linked organization, Concept of circular queue, double ended queue, Applications of queue: priority queue.	8
4	Trees Tree: Trees and binary trees-concept and terminology, Expression tree, Binary tree as an ADT, Binary search tree, Recursive and Non recursive algorithms for binary tree traversals ,Binary search tree as ADT(Insert Search Delete, level wise Display), Types of Trees: OBST, AVL Trees , Threaded binary Tree: Concept of threaded binary tree (inorder, preorder and postorder). Preorder and In-order traversals of in-order threaded binary tree ,Heap- Heap data structure ,Min and Max Heap, Heap Sort	8
5	Graph Concept and terminologies, Graph as an ADT, Representation of graphs using adjacency matrix and adjacency list, Breadth First Search Traversal, Depth First Search Traversal, Prim's and Kruskal's algorithm for minimum spanning tree, Shortest path using Dijkstra's algorithm, topological sorting.	7
Total Hours		39

List of Experiments

1. Consider a student database of SEIT class (at least 15 records). Database contains different fields of every student like Roll No, Name and SGPA.(array of structure) a) Design a roll call list, arrange list of students according to roll numbers in ascending order (Use Bubble Sort)	4 Hrs
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b) Arrange list of students alphabetically. (Use Insertion sort) c) Arrange list of students to find out first ten toppers from a class. (Use Quick sort) d) Search students according to SGPA. If more than one student having same SGPA, then print list of all students having same SGPA. (use Linear Search) e) Search a particular student according to roll number using binary search without recursion	
2. Implement stack as an abstract data type using singly linked list and use this ADT for conversion of Infix expression to postfix expression and postfix expression evaluation OR Infix expression to prefix expression and prefix expression evaluation	2 Hrs
3. Implement Circular Queue using Array. Perform following operations on it. a) Insertion (Enqueue) b) Deletion (Dequeue) c) Display (Note: Handle queue full condition by considering a fixed size of a queue.)	2 Hrs
4. Construct an Expression Tree from postfix or prefix expression. Perform recursive and non-recursive In-order, Pre-order and Post-order traversals	4 Hrs
5. Implement binary search tree and perform following operations: a) Insert (Handle insertion of duplicate entry) b) Delete c) Search d) Display tree (Level-wise, print total count of leaf nodes)	4 Hrs
6. Implement In-order Threaded Binary Tree and traverse it in In-order and Pre-order	2 Hrs
7. Represent a graph of your college campus using adjacency list /adjacency matrix. Nodes should represent the various departments/institutes and links should represent the distance between them. Find minimum spanning tree a. Prim's Algorithm Kruskal's Algorithm	4 Hrs
8. Represent a graph of city using adjacency matrix /adjacency list. Nodes should represent the various landmarks and links should represent the distance between them. Find the shortest path using Dijkstra's algorithm from single source to all destination.	4 Hrs
9. Implementation of Hash table using array and handle collisions using a. Linear probing with and without replacement Chaining with and without replacement	4 Hrs
Total Hours	30
Guideline for Instructor: Experiments to be conducted in C Language	
Text Books	
1. Gilberg, B. Forouzan, "Data Structures: A pseudocode approach with C", Cengage Learning, 2nd edition. 2. Alfred Aho, John Hopcroft, Jeffrey Ullman, "Data Structures & Algorithms", Pearson Publication 3. Y. Kanetkar, "Data Structures through C," 5th edition BPB Publications, 2023	
Reference Book	
1. Reema Thareja, "Data Structures using C ", Second Edition, Oxford Higher Publication 2. E. Horowitz, S. Sahani, "Fundamentals of Data Structures in C", Universities press, 2 nd edition 3. Jean-Paul Tremblay & Paul G. Sorenson, "An Introduction to Data Structures with Applications" Publisher-Tata McGraw Hill, 2nd edition 4. Data Structures using C & C++-By Tanenbaum Publisher-Prentice-Hall International 5. Fundamentals of Computer Algorithms by Horowitz , Sahani, Galgotia Pub 2001 ed.	
Online References	
1. https://nptel.ac.in/courses/106106133 2. https://nptel.ac.in/courses/106102064 3. https://www.coursera.org/specializations/data-structures-algorithms	

Second Year B. Tech Information Technology														
Autumn Semester														
Course Code: B25IT302					Course Name: Object Oriented Programming									
Teaching Scheme (Hours/Week)					Examination Scheme						Credits			
L	P	T	OL	ODL	CIE	ETE	TW	PR	OR	TOTAL	L	P	T	TOTAL
3	2	-	-	-	40	60	-	50	-	150	3	1	-	4
Course Outcomes: After completing the course, the students will be able to: CO1: Demonstrate the Object-Oriented Programming concept CO2: Design and develop programs using classes, constructors, and destructors to represent real-world entities. CO3: Apply inheritance to create derived classes from base classes for code reusability CO4: Implement file input and output (I/O) operations in C++ CO5: Apply built-in algorithms from STL to efficiently solve common problems.														
Unit	Contents													Hrs.
1	Basic of OOP: Principles of Object Oriented Programming (OOP), features of OOP. Basics of C++: Characteristics of C++, Structure of a C++ Program, Writing and Running C++ Programs. C++ Fundamentals: Variables, Data Types, Constants, Operators in C++, Expressions and Statements, Input/output in C++, Control Structures (if, for, while, etc.),function, string.													7
2	Class and Object: Creating class and Objects, Member function, array of object, Access Modifiers: public, private, and protected, static variable, inline function. Constructors and Destructors: Introduction to Constructors, Default Constructors and Parameterized Constructors, Destructor Functions, Dynamic Memory Allocation.													8
3	Inheritance and Polymorphism: Inheritance: Basic Concepts, types, Constructor and Destructor in Inheritance, Access Control in Inheritance. Ambiguity in Multiple Inheritance, Virtual Base Class, Abstract class. Polymorphism: Introduction of Polymorphism, Function Overloading, Function Overriding. Operator Overloading: Introduction, syntax, overloading unary and binary operators. Runtime Polymorphism: Pointers to Base class, virtual function and its significance in C++, pure virtual function and virtual table, virtual destructor, abstract base class.													8
4	Friend Functions: Friend Function Definition, Friend Classes. File Handling: File Input and Output (I/O), File Operations, Reading from and Writing to Files, File Handling Classes in C++: fstream, ifstream, and ofstream classes													8
5	Exception Handling: Fundamentals, other error handling techniques, simple exception handling- Divide by Zero, Multiple catching, re-throwing an exception, exception specifications, user defined exceptions, processing unexpected exceptions Templates & STL: Function and Class templates and using STL like containers, algorithms													8
													Total Hours	39

List of Experiments		
1	Write a program based on basic control structure in C++	2 Hrs
2	Create a simple application with classes. Use array of object and access modifiers	4 Hrs
3	Design and develop the program for a given problem statement. Demonstrate the constructor and destructor in program	4 Hrs
4	Design and develop inheritance concepts for a given case study, identify objects and relationships and implement inheritance wherever applicable.	4 Hrs
5	Design and Develop the program to demonstrate the concept of static and run time polymorphism.	2 Hrs
6	Design and develop the program using the concept of Operator overloading. (at least two Unary and binary operator)	2 Hrs
7	Implement a program for maintaining a database of given case study using Files.	4 Hrs
8	Implement a program using the template for given problem statement	4 Hrs
9	Implement a program to demonstrate various types of Exception.	2 Hrs
10	Write C++ program using STL for sorting and searching user defined records such as personal records (Name, DOB, Telephone number etc) using vector containers.	4 Hrs
11	Write a program using STL containers.	2 Hrs
Total		34 Hrs
Text Books		
1. Robert Lafore; 'Object-Oriented Programming in C++', Fourth Edition, Sams Publishing, ISBN:0672323087		
2. E.Balagurusamy; 'Object-Oriented Programming with C++' 7th edition, Graw-Hill Publication, ISBN 10: 9352607996 ISBN 13: 9789352607990		
3. Deitel, 'C++ How to Program', 4th Edition, Pearson Education, ISBN:81-297-0276-2		
Reference Book		
1. Bjarne Stroustrup ; 'The C++ Programming Language', Fourth Edition, Addison-Wesley		
2. Bjarne Stroustrup ; 'A Tour of C++' , Second Edition, Addison-Wesley		
3. Herbert Schildt, 'C++-The complete reference', Eighth Edition, McGraw Hill Professional, 2011, ISBN:978-00-72226805		
4. Matt Weisfeld, 'The Object-Oriented Thought Process', Third Edition,Pearson ISBN-13:075- 2063330166		
5. Cox Brad, Andrew J. Novobilski, 'Object –Oriented Programming: An Evolutionary Approach, Second Edition, Addison–Wesley, ISBN:13:978-020-1548341		
Online References		
1. Programming in C++: https://onlinecourses.nptel.ac.in/noc21_cs02/preview		
2. Programming in Modern C++: https://onlinecourses.nptel.ac.in/noc22_cs43/preview		
3. https://www.linkedin.com/learning/c-plus-plus-standard-template-library		
4. https://www.linkedin.com/learning/c-plus-plus-essential-training-15106801/		

Second Year B. Tech Information Technology															
Autumn Semester															
Course Code : B25IT303					Course Name: Design Thinking for User Experience: UI/UX										
Teaching Scheme (Hours/Week)					Examination Scheme						Credits				
L	P	T	OL	ODL	CIE	ETE	TW	PR	OR	TOTAL	L	P	T	TOTAL	
3	-	-	-	-	40	60		-	-	100	3	-	-	3	
Course Outcomes: After completing the course, the students will be able to: CO1: Apply the principles of User Interface design. CO2: Apply the fundamentals of user experience. CO3: Evaluate strategies for managing design projects. CO4: Design solutions to address the challenges associated with information visualization. CO5: Evaluate usability of a user-interface design.															
Unit	Contents														Hrs.
1	Introduction and Overview of UI: The Human -I/P, O/P channels, Human Memory, thinking, emotion, individual difference (diversity), human psychology. The Relationship Between UI and UX , Roles in UI/UX, Interface Conventions, Approaches to Screen Based UI, Template vs Content, Formal Elements of Interface Design, Active Elements of Interface Design, Composing the Elements of Interface Design, UI Design Process, Visual Communication design component in Interface Design , Application of UI design Introduction to Design Technologies and Tools Sketch ,Wireframe ,Invision, Axure, Figma, Flutter, Mockups														8
2	User Experience: Foundation of UX design, Good and poor design, Understanding Your Users, Designing the Experience Elements of user Experience, Visual Design Principles, Functional Layout, Interaction design, Introduction to the Interface, Navigation Design, User Testing, Developing and Releasing Your Design User experience and user interaction-Usability of interactive systems, goals and measures, Universal Usability, Characteristics of graphical and web user interfaces, guidelines, principles and theories of good design, User Experience-Concept of UX, Trends in UX, 6 Stages used to UX design , Applications of UX design														8
3	Design Process: Principles that support usability, Design standards, Design Guidelines, Interaction design, The software design process, User focus, Scenarios, Navigation Design, Screen Design, Prototyping techniques, Wire-Framing, Understanding the UI Layer and Its Execution Framework, Model-View-Controller(MVC) Framework, Interaction Styles														8
4	Implementation support and Screen Based Controls: Implementation support: Support, training and learning, requirement of user support, element of windowing systems, Individual window design, multiple window design, command organization strategies command menus, natural languages in computer Screen Based Controls: Selection control-Radio buttons, check boxes, list boxes, Read-only controls- text boxes, Operable controls - buttons, slider, tab, scroll bar, clear text and messages, text for web pages, Graphics, icons and images, Presentation controls-Static text fields, Group boxes, column headings, tool tips, progress indicators														8
5	Evaluation Techniques: User interface management system (UIMS), Goals of evaluation, Categorization of Evaluation techniques, Choosing an Evaluation Method. DECIDE, Heuristic Evaluation, cognitive walkthrough, Usability testing														7
Total Hours														39	
List of Activities															
1. Designing a Responsive User Interface for a Mobile App 2. Designing Navigation for a Multi-Page Website 3. Apply the MVC Framework in a Design Task 4. Create a Multi-Window Application with Effective Screen-Based Controls															

5. Evaluate the Usability of a Mobile or Web Application
Text Books
1. Creative Tim, "Fundamentals of Creating a Great UI/UX", First Edition 2. Jon Yablonski, "Laws of UX: Using Psychology to Design Better Products & Services", O'Reilly Media, Inc., 21-Apr-2020, First Edition 3. Jenifer Tidwell, Charles Brewer, Aynne Valencia "Designing Interfaces: Patterns for Effective Interaction Design", O'Reilly Media, Inc., First Edition. 4. Designing The User Interface: Strategies for Effective Human-Computer Interaction. Pearson Education Limited. ISBN 978-1-292-03701-1. 5. Alan Dix (2008). Human Computer Interaction. Pearson Education. ISBN 978-81-317-1703-5
Reference Book
1. Shneiderman, Plaisant, Cohen, Jacobs, "Designing the User Interface-Strategies for Effective Human Computer Interaction", 5th Edition, PEARSON Publication, ISBN 97881317-3255-7 2. Wilbert O. Galitz "The Essential Guide to User Interface Design", 2 nd Edition, WILEY Publication, 9780471271390, 047127139X. 3. Alan Dix, Janet Finlay, Gregory D. Abowd, Russell Beale, "Human-Computer Interaction, 3rd Edition, PEARSON education, 9788131717035, 8131717038 4. Alan Coopen, "The essentials of interaction", Wiley, ISBN:9781568843223, 1568843224 5. Gerard Jounghyun Kim (20 March 2015). Human-Computer Interaction: Fundamentals and Practice. CRC Press. ISBN 978-1-4822-3390-2. 6. Donald A. Norman (2013). The Design of Everyday Things Basic Books. ISBN 978-0-465-07299-6
Online References
1. "The Guide to Wireframing" by UXPin: https://www.uxpin.com/studio/ebooks/guide-to-wireframing/ This eBook provides an in-depth guide to wireframing, covering the basics, best practices, and tips for creating effective wireframes. 2. "UX Design for Startups" by Marcin Tredar:: https://uxpin.com/studio/ebooks/ux-design-forstartups/ - This eBook focuses on UX design principles and strategies specifically tailored for startups, covering topics like user research, prototyping, and user testing. 3. https://hcibib.org/ 4. https://onlinecourses.nptel.ac.in/noc21_ar05/preview

Second Year B. Tech Information Technology															
Autumn Semester															
Course Code: B25IT304					Course Name: Experiential Learning Courses: Project Based Learning										
Teaching Scheme (Hours/Week)					Examination Scheme						Credits				
L	P	T	OL	ODL	CIE	ETE	TW	PR	OR	TOTAL	L	P	T	TOTAL	
-	4	-	-	-	-	-	50	-	-	50		2	-	2	
Course Outcomes: After completion of the course, student will be able to - CO1: Design solution to real life problems and analyze its concerns through shared cognition. CO2: Apply learning by doing an approach in PBL to promote lifelong learning. CO3: Tackle technical challenges for solving real world problems with team efforts. CO4: Collaborate and engage in multi-disciplinary learning environments.															
Course Framework															
1	Group Structure: Working in supervisor/mentor monitored groups; the students plan, manage, and complete a Task/project/activity which addresses the stated problem. 1. There should be team/group of 3-4 students 2. A supervisor/mentor teacher assigned to individual groups														
2	Selection of Project/Problem: The problem-based project oriented model for learning is recommended. The model begins with the identifying a problem, often growing out of a question or “wondering”. This formulated problem then stands as the starting point for learning. Students design and analyze the problem within an articulated interdisciplinary or subject frame. A problem can be theoretical, practical, social, technical, symbolic, cultural and/or scientific and grows out of students’ wandering within different disciplines and professional environments. A chosen problem has to be exemplary. The problem may involve an interdisciplinary approach in both the analysis and solving phases .By example, a problem needs to refer back to a particular practical, scientific, social and/or technical domain. The problem should stand as one specific example or manifestation of more general learning outcomes related to knowledge and/or modes of inquiry. There are no commonly shared criteria for what constitutes an acceptable project. Projects vary greatly in the depth of the questions explored, the clarity of the learning goals, the content and structure of the activity. • A few hands-on activities that may or may not be multidisciplinary • Use of technology in meaningful ways to help them investigate, collaborate, analyze, synthesize and present their learning. • Activities may include-Solving real life problems, investigation /study and Writing reports of in depth study, field work.														
3	Assessment: The institution/head/mentor is committed to assessing and evaluating both student performance and program effectiveness. Progress of PBL is monitored regularly on a weekly basis. Weekly review of the work is necessary. During the process of monitoring and continuous assessment AND evaluation the individual and team Performance is to be measured. PBL is monitored and continuous assessment is done by supervisor/mentor and authorities. Students must maintain an institutional culture of authentic collaboration, self-motivation, peer-learning and personal responsibility. The institution/department should support students in this regard through guidance/orientation programs and the provision of appropriate resources and services. Supervisor/mentor and Students must actively participate in assessment and evaluation processes. Group may demonstrate their knowledge and skills by developing a public product and/or report and/or presentation. • Individual assessment for each student (Understanding individual capacity, role and involvement in the project) • Group assessment (roles defined, distribution of work, intra-team communication and togetherness) Documentation and presentation Evaluation and Continuous Assessment: It is recommended that all activities are to be recorded and regularly, regular assessment of work to be														

	done and proper documents are to be maintained at college end by both students as well as mentors (you may call it PBL work book). Continuous Assessment Sheet (CAS) is to be maintained by all mentors/departments and institutes. Recommended parameters for assessment, evaluation and weightage: • Idea Inception (5%) • Outcomes of PBL/ Problem Solving Skills/ Solution provided/ Final product (60%) (Individual assessment and team assessment) • Documentation (Gathering requirements, design & modeling, implementation/execution, use of technology and final report, other documents) (10%) • Demonstration (Presentation, User Interface, Usability etc) (10%) • Contest Participation/ publication (10%) • Awareness /Consideration of -Environment/ Social /Ethics/ Safety measures/Legal aspects (5%)
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Second Year B. Tech Information Technology														
Autumn Semester														
Course Code: B24IT305					Course Name: Logic Design & Computer Organization									
Teaching Scheme(Hours/Week)					Examination Scheme						Credits			
L	P	T	OL	ODL	CIE	ETE	TW	PR	OR	TOTAL	L	P	T	TOTAL
2	2	-	-	-	40	60	25	-	-	125	2	1	-	3
Course Outcomes: After completing the course, the students will be able to: CO1: Recognize the pin layout of the several digital integrated circuits used in the laboratory. CO2: Conduct the experiment and verify the properties of various logic circuits. CO3: Analyze the sequential and combinational circuits. CO4: Design any sequential or combinational circuit using Hardware CO5: Identify basic components and design of the CPU: the ALU and control unit and compare and select various Memory devices as per requirement.														
Unit	Contents													Hrs.
1	Introduction: Digital signals, digital circuits, AND, OR, NOT, NAND, NOR and Exclusive-OR operations, Boolean algebra, examples of IC gates, Number Systems: binary, signed binary, octal hexadecimal number, binary arithmetic, one's and two's complements arithmetic, codes, error detecting and correcting codes													5
2	Combinational Logic Design : Standard representation for logic functions, K-map representation, simplification of logic functions using K-map, minimization of logical functions, Don't care conditions, Multiplexer, Demultiplexer or Decoders, Adders, Subtractors, BCD arithmetic, carry look ahead adder, serial adder, ALU, elementary ALU design, parity checker or generator.													5
3	Sequential circuits and systems : 1-bit memory, the circuit properties of Bistable latch, the clocked SR flip flop, J-K-T and D-types flip flops, applications of flip flops, shift registers, applications of shift registers, serial to parallel converter, parallel to serial converter, ring counter, sequence generator, ripple (Asynchronous) counters, synchronous counters, counters design using flip flops, special counter IC's, asynchronous sequential counters, applications of counters.													5
4	Basic Computer Organization and Design: Instruction codes, Computer Registers, Computer Instructions and Instruction cycle. Timing and Control, Memory-Reference Instructions, Input-Output and interrupt. Central processing unit: Stack organization, Instruction Formats, Addressing Modes, Data Transfer and Manipulation, Complex Instruction Set Computer (CISC) Reduced Instruction Set Computer (RISC), CISC vs RISC.													5
5	Register Transfer and Micro Operations: Register Transfer Language, Register Transfer, Bus and Memory Transfers, Arithmetic Micro-Operations, Logic Micro-Operations, Shift Micro-Operations, Arithmetic logic shift unit. Micro-Programmed Control: Control Memory, Address Sequencing, Micro-Program example, Design of Control Unit. Memory System: Memory Hierarchy, Semiconductor Memories, RAM(Random Access Memory), Read Only Memory (ROM), Types of ROM, Cache Memory, Performance considerations, Virtual memory, Paging, Secondary Storage, RAID. Input Output: I/O interface, Programmed IO, Memory Mapped IO, Interrupt Driven IO, DMA													6
Total Hours													25	
List of Experiments														

1	• Verification of truth tables of the following Logic gates Two input (i) OR (ii) AND (iii) NOR (iv) NAND (v) Exclusive-OR (vi) Exclusive-NOR • Design a simple combinational circuit with four variables and obtain minimal expression and verify the truth table using Digital Trainer Kit	2
2	• Verification of functional table of 3 to 8-line Decoder or De-multiplexer. • 4 variable logic function verification using 8 to 1 multiplexer.	2
3	• Design a full adder circuit and verify its functional table. • Design a full subtractor circuit and verify its functional table.	2
4	Draw the circuit diagram of a single bit comparator and test the output Construct 7 Segment Display Circuit Using Decoder and 7 Segment LED and test it.	2
5	Draw the circuit diagram of MOD-8 ripple counter and construct a circuit using T-Flipflops and Test.	2
6	Design MOD-8 synchronous counter using T Flip-Flop and verify the result and sketch the output waveforms.	2
7	Verification of functional tables of (i) JK Edge triggered Flip-Flop (ii) JK Master Slave Flip-Flop (iii) D Flip-Flop	2
8	Design a four-bit ring counter using D Flip-Flops or JK Flip Flop and verify output	2
9	Simulate the 4 bit ALU using 1 bit ALU using V lab	2
10	Simulation of 8 Bit RAM using V lab	2
Total Hours		20
Text Books		
1. R. P. Jain, Modern Digital Electronics, McGraw Hill Education, 2009. 2. M. Moris Mano (2006), Computer System Architecture, 3rd edition, Pearson/PHI, India. 3. William Stallings (2010), Computer Organization and Architecture- designing for performance, 8th edition, Prentice Hall, New Jersey.		
Reference Book		
1. M. M. Mano, Digital logic and Computer design, Pearson Education India, 2016. 2. Kumar, Fundamentals of Digital Circuits, Prentice Hall India, 2016. 3. Douglas Hall, Microprocessors and Interfacing, McGraw-Hill Publication, Revised 2nd Edition, 2006. 4. Carl Hamacher, Zvonks Vranesic, SafeaZaky (2002), Computer Organization, 5th edition, McGraw Hill, New Delhi, India. 5. Anrew S. Tanenbaum (2006), Structured Computer Organization, 5th edition, Pearson Education Inc 6. John P. Hayes (1998), Computer Architecture and Organization, 3rd edition, Tata McGrawHill		
Online References		
1. http://vlabs.iitkgp.ac.in/coa/index.html 2. https://onlinecourses.nptel.ac.in/noc21_cs61/preview 3. https://onlinecourses.nptel.ac.in/noc20_cs67/preview		

Second Year B. Tech Information Technology														
Spring Semester														
Course Code: B25IT306					Course Name: Problem Solving and Analytical Skills									
Teaching Scheme (Hours/Week)					Examination Scheme						Credits			
L	P	T	OL	ODL	CIE	ETE	TW	PR	OR	TOTAL	L	P	T	TOTAL
-	-	2	-	-	-	-	50	-	-	50	-	-	2	2
Course Outcomes: After completing the course, the students will be able to: CO1: Understand the basic concepts of arithmetic ability, business computations and data interpretation. CO2: Learn to develop strategic approaches to problem solving. CO3: Apply mathematical and statistical methods to analyze real-world problems. CO4: Develop logical reasoning and thinking ability. CO5: Develop skills to meet the competitive examinations for better job opportunity.														
Unit	Contents													Hrs.
1	Arithmetic ability: Algebraic operations BODMAS, Fractions, LCM & GCD(HCF), Probability, Permutations and Combinations Verbal Reasoning: Number Series, Coding & Decoding, Blood relationship, Clocks, Calendars.													7
2	Quantitative aptitude: Averages, Ratio and proportion, Time-distance. Business computations: Percentages, Profit & loss, Simple compound interest, Stocks and Shares.													7
3	Data Interpretation: Tabulation, Bar Graphs, Pie Charts. Logical Venn Diagrams, Tabulation													7
4	Non-verbal reasoning: Series, Analogy, Problems on Cube and Dice, Mirror image, Water image, Rule Detection, Analytical reasoning													7
Total Hours													28	
List of Activities														
1	I. Solve problems that require finding the LCM and GCD of large numbers. II. Group Quiz: Mix arithmetic and reasoning problems for team competitions.													
2	I. Solve basic ratio and proportion problems. II. problems on profit and loss involving cost price, selling price, and profit percentage. III. Solve problems on dividend, face value, and market value of stocks.													
3	Case Study: Provide a real-world scenario (e.g., financial data, election results) and ask students to interpret it using tabulation, bar graphs, and pie charts.													
4	Create your own figure series using shapes (circle, square, triangle) and ask a peer to find the next figure in your series.													
5	Draw your own cube net, label the faces, and fold it into a cube. Share your folded cube with the class and ask your peers to identify opposite and adjacent faces.													
Online References														
1. Quantitative Aptitude for Competitive Examination by R S Agrawal, S.Chand publications. 2. https://www.lpude.in/SLMs/Master%20of%20Computer%20Applications/Sem_1/DEPEA515_ANALYTICAL_SKILLS-I.pdf 3. Quantitative Aptitude and Reasoning by R V Praveen, PHI publishers.														

Second Year B. Tech Information Technology															
Autumn Semester															
Course Code :B25IT307					Course Name: Professional Communication and Presentation Skill										
Teaching Scheme (Hours/Week)					Examination Scheme						Credits				
L	P	T	OL	ODL	CIE	ETE	TW	PR	OR	TOTAL	L	P	T	TOTAL	
-	-	2	-	-	-	-	50	-	-	50	-	-	2	2	
Course Outcomes : After completing the course, the students will be able to: CO1: Demonstrate effective rhetoric, persuasive skills, audience engagement, and adaptability in communication CO2: Apply visual aids, body language, voice modulation, and confidently manage audience questions and maintain engagement throughout. CO3: Analyze information and construct logical arguments to solve problems and make informed decisions. CO4: Exhibit professional and technical writing skills and avoid common pitfalls, ensuring clear, polite, and informative communication.															
Unit	Contents													Hrs.	
1	Public Speaking: Using rhetoric and understanding the context, controlling nervousness, building confidence, stage presence, poise, and impact, Voice modulation, pace, volume & pitch Activities: Two Truths and a Lie, Story telling													6	
2	Presentation Skills: Structuring a presentation & using visual aids , Audience connection strategies & feedback, Handling Q&A session with confidence. Activities: Idea Presentation, Advertisement Creation													6	
3	Critical Thinking: Case Studies and Situational Analysis, Structuring the argument and handling rebuttal, Understanding logical fallacies and cognitive biases. Activities: Case Study, debate.													6	
4	Technical Writing: Structuring emails for different purposes (requests, complaints, updates, and follow-ups) , avoiding common mistakes and the right use of CC, BCC, and reply-all etiquettes, Blog/article/ newsletter format and examples, Utilizing the information through the right resources Activities: EMail writing (Professional & Personal), Blog writing, Survey Reports.													6	
5	Interview Skills: Common interview questions and effective responses using the STAR (Situation, Task, Action, Result) technique, Non-verbal cues: eye contact, posture, gestures and attire, Mock interviews and role-playing exercises. Activities: Group discussion, Fishbowl Discussion.													6	
Total Hours													30		
List of Tutorials															
1	Two Truths and a Lie: Each player shares three statements about themselves. Two of these statements are true, and one is a lie. The other players try to guess which statement is the lie. It encourages active listening, critical thinking, and engaging conversation.														
2	Storytelling: Storytelling is the art of conveying ideas, emotions, and experiences through structured narratives. It helps in simplifying complex technical concepts, making them more relatable and easier to understand. Additionally, storytelling enhances communication skills, enabling students to effectively present their ideas and collaborate with others in the tech industry.														
3	Idea Presentation: An idea presentation is a concise and compelling delivery designed to introduce a new concept, product, or service to an audience. It's a crucial skill in various settings, from business meetings to academic conferences. Students will be presenting for a Seminar/Presentation and handling Q&A sessions.														
4	Advertisement Creation: Creating a successful advertisement involves a multi-faceted approach that considers the target audience, the product/service, and the desired message. Students will create an														

	advertisement and present it in class.
5	Case Study Analysis: A case study analysis involves a deep dive into a specific situation or event to understand its complexities, identify underlying causes, and draw relevant conclusions.
6	Debate: A debate is a formal discussion on a particular topic or collection of topics, often with a moderator and an audience. In a debate, arguments are put forward for opposing viewpoints.
7	EMail writing(Professional & Personal): Write emails : requests, complaints, updates, and follow-ups.
8	Blog writing : Blogging is a powerful tool for sharing your thoughts, ideas, and experiences with the world. Students will write a blog/article/ newsletter/survey report.
9	Survey Reports: A survey report is a document that presents the findings of a survey in a clear, concise, and organized manner. It translates raw data into actionable insights that can inform decision-making. Students will draft one such report.
10	Group discussion: A group discussion is a structured conversation where a group of individuals engage in a focused exchange of ideas and perspectives on a particular topic. It's a valuable skill in various settings, from academic discussions to workplace meetings. Students will engage in GDs based on relevant topics.
Text Books	
1. "The Art of Public Speaking" by Stephen E. Lucas 2. "Corporate Communication: Principles and Practices" by Jaishri Jethwaney and Sanjay Bhargava 3. "Critical Thinking and Problem Solving" by Dr. Ranjit Singh 4. "Essentials of Business Communication" by Rajendra Pal and J. S. Korlahalli 5. "Competency-Based Interviews" by Robin Kessler	
Reference Book	
1. "You Can Win" by Shiv Khera 2. "Talk Like TED" by Carmine Gallo 3. "Presentation Zen: Simple Ideas on Presentation Design and Delivery" by Garr Reynolds 4. "Think Critically" by Peter Facione and Carol Ann Gittens 5. "The Power of Critical Thinking" by Lewis Vaughn 6. "The Etiquette Advantage in Business: Personal Skills for Professional Success" by Peggy Post and Peter Post 7. "How to Win at Interviews" by Pramila Mathew 8. "Cracking the Code to a Successful Interview: 15 Insider Secrets from a Top-Level Recruiter" by Evan Pellett	
Online References	
1. Technical English for Engineers - Course 2. English for Research Paper Writing - Course 3. Personality Development and Communication Skills - Course	

Second Year B. Tech Information Technology														
Spring Semester														
Course Code : B25IT401					Course Name: Database Management Systems									
Teaching Scheme(Hours/Week)					Examination Scheme						Credits			
L	P	T	OL	ODL	CIE	ETE	TW	PR	OR	TOTAL	L	P	T	TOTAL
3	2	-	-	-	40	60		-	50	150	3	1	-	4
Course Outcomes : After completing the course, the students will be able to: CO1: Analyze and design Database Management System using ER model CO2: Implement database queries using SQL CO3:Normalize the database design using normal forms CO4:Apply Transaction Management concepts in real-time situations CO5:Use NoSQL databases for processing unstructured data														
Unit	Contents													Hrs.
1	Introduction to Database System and ER Modeling: Introduction to Database Management Systems, Purpose of Database Systems, Database-System Applications, View of Data, Database Languages, Database System Structure, Data Models, Database Design and ER Model: Entity, Attributes, Relationships, Constraints, Keys, Design Process, Entity Relationship Model, ER Diagram, Design Issues, Extended E-R Features, converting E-R & EER diagram into tables. Case Study: Design ER Model for any real time application and convert the same into tables on paper.													8
2	SQL: Characteristics and advantages, SQL Data Types, DDL, DML, DCL, TCL, SQL Operators, Views: Creating, Dropping, Updating using Indexes, Set Operations, Predicates and Joins, Set membership, Tuple Variables, Set comparison, Ordering of Tuples, Aggregate Functions, SQL Functions, Nested Queries, Triggers.													8
3	Relational Database Design: Relational Model: Basic concepts, Attributes and Domains, CODD's Rules, Relational Integrity: Domain, Referential Integritys, Enterprise Constraints, Database Design: Features of Good Relational Designs, Normalization, Atomic Domains and First Normal Form, 2NF, 3NF, BCNF. Case Study: Convert a Relational Database to different Normal Forms.													8
4	Database Transaction Management: Transaction concept, Transaction states, ACID properties, Concept of Schedule, Serial Schedule, Serializability: Conflict and View, Cascaded Aborts, Recoverable and Non-recoverable Schedules, Concurrency Control: Lock-based, Time-stamp based Deadlock handling, Recovery methods: Shadow- Paging and Log-Based Recovery													8
5	Advances in Databases and Big Data: Introduction to Big Data : 3 V's of Big Data, Solution for Big Data Introduction to NoSQL databases, NoSQL data models and MongoDB, CAP theorem, BASE properties, and comparative study of SQL and NoSQL													7
Total Hours													39	
List of Experiments														
1	Install and configure client and server of MySQL. (Show all commands and necessary steps for installation and configuration)													2
2	Decide a real time application and formulate a problem statement for the application to be developed. Propose a Conceptual Design using ER features using tools like ERD plus, ER Win etc. (Identifying entities, relationships between entities, attributes, keys, cardinalities, generalization, specialization etc.) Convert the ER diagram into a table.													2
3	Design and Develop SQL queries for DDL statements for suitable database application.													2
4	Design SQL queries for suitable database application using SQL DML statements:Insert, Select,													2

	Update, delete with operators, functions, Set operators	
5	Implement arithmetic operators, logical operators, pattern matching, IN and NOT IN predicates, and MySQL built-in functions.	4
6	Implement SQL queries for different SQL concepts such as grouping data, using the GROUP BY clause, employing the HAVING clause, applying the EXISTS/NOT EXISTS operators, Creating and using Database Views	4
7	Write SQL queries to demonstrate the different SQL concepts like subqueries, performing various join operations, and using set operators in MySQL.	4
8	Implement SQL queries to demonstrate the nested queries in SQL using MySQL.	2
9	Design and Develop MongoDB Queries using CRUD operations. (Use CRUD operations, SAVE method, logical operators etc)	2
10	MongoDB - Aggregation and Indexing: Design and Develop MongoDB Queries using aggregation and indexing with suitable examples using MongoDB.	4
11	MongoDB - Map reduce operations: Implement Map reduces operation with a suitable example using MongoDB.	2
Total Hours		30
Text Books		
1. Silberschatz A., Korth H., Sudarshan S., "Database System Concepts", McGraw Hill Publishers, 7th Edition, 2020 ISBN 978-0-07-802215-9. 2. Ivan Bayross, "SQL, PL/SQL the Programming Language of Oracle", BPB Publications, 2014 ISBN: 9788176569644. 3. Connally T, Begg C., "Database Systems- A Practical Approach to Design, Implementation and Management", Pearson Education, 5th Edition, 2010, ISBN 81-7808-861-4. 4. Pramod J. Sadalage and Martin Fowler, "NoSQL Distilled", Addison Wesley, ISBN 10: 0321826620, 2013, ISBN 13: 978-0321826626.		
Reference Book		
1. C. J. Date, "An Introduction to Database Systems", Addison-Wesley, 8th Edition, 2004, ISBN 0321189566. S. K. Singh, "Database Systems: Concepts, Design and Application", Pearson Education, 2009, ISBN 9788177585674. Kristina Chodorow, Michael Dierolf, "MongoDB: The Definitive Guide", O'Reilly Publications, 3rd Edition, 2019 ISBN 9781491954461. 2. Kevin Roebuck, "Storing and Managing Big Data - NoSQL, HADOOP and More", Emereo Pty Limited, 2011, ISBN 1743045743, 9781743045749.		
Online References		
1. https://dev.mysql.com/doc/ 2. https://www.coursera.org/learn/database-management 3. https://archive.nptel.ac.in/courses/106/105/106105175/		

Second Year B. Tech Information Technology															
Spring Semester															
Course Code: B25IT402										Course Name: Operating System					
Teaching Scheme (Hours/Week)					Examination Scheme						Credits				
L	P	T	OL	ODL	CIE	ETE	TW	PR	OR	TOTAL	L	P	T	TOTAL	
3	2	-	-	-	40	60	-	50	-	150	3	1	-	4	
Course Outcomes:															
After completing the course, the students will be able to:															
CO1: Understand the objectives, architecture, and basic commands of operating systems, with a focus on Linux OS and shell scripting.															
CO2: Apply process scheduling and synchronization techniques, including thread programming, to optimize CPU utilization and manage concurrency.															
CO3: Analyze process synchronization methods and deadlock handling techniques to ensure efficient and deadlock-free operations.															
CO4: Evaluate memory management techniques and I/O scheduling methods for effective resource utilization in modern operating systems.															
CO5: Design secure operating system environments by implementing protection mechanisms and monitoring system threats.															
Unit	Contents													Hrs.	
1	Overview Of Operating System: Operating System Objectives, The Evolution of Operating System, OS Design Considerations for Multiprocessor and Multicore OS, Architecture of Operating System: Monolithic, Microkernel, Exokernel, Introduction to Linux OS, Basic Shell Commands, Shell Scripting using BASH.													7	
2	Process Management : Process: Process Concept, Process States, Process Control Block, Process Description Threads: Process and Threads, Basic types of threads, Multithreading, Thread Programming using thread library APIs Process Scheduling: Types of Scheduling, Scheduling Criteria, Scheduling Algorithms: First-Come First- Served, Shortest-Job-First, Priority, Round Robin, Case Study - Linux Scheduling													8	
3	Process Synchronization : Principles of Concurrency, Critical - Section Problem, Mutual Exclusion: Requirements, Operating System support - Semaphore and Mutex, Classical Synchronization Problems: Reader-Writer Problem, Producer - Consumer Problem, Real Life Problems, Inter-Process Communication: Pipes and Shared Memory Deadlock: Principles of Deadlock, Deadlock Characterization: Necessary Conditions, Resource - Allocation Graph, Methods for Handling Deadlock: Deadlock Prevention, Deadlock Avoidance, Deadlock Detection and Recovery, Case Study: Dining Philosopher Problem													8	
4	Memory and I/O Management: Memory Management Requirements, Memory Partitioning: Fixed Partitioning, Dynamic Partitioning, Buddy System, Paging, Segmentation, Virtual Memory: Demand Paging, Page Replacement, Thrashing I/O Management: I/O Devices, Organization of the I/O Function, I/O Buffering, Secondary Storage Management: Disk Structure, Disk Scheduling File Management: Overview-Files and File Systems, File structure. File Organization and Access, File Directories, File Sharing, Case Study: Linux File System, Android File System Case Study: Linux Operating System													9	
5	Protection And Security: Goals of protection, Domain of protection, Access matrix, Implementation of access matrix, Revocation of access rights, Security problems, Authentication, Program threats, System threats, Threat monitoring.													7	
Total Hours													39		

List of Experiments (Minimum 10 experiments to be performed)		
1	Study of Basic Linux Commands: echo, ls, read, cat, touch, test, loops, arithmetic comparison, conditional loops, grep, sed find, diff, tac and Bash scripting	2
2	Implement the C program in which the main function accepts the integers to be sorted. Main function uses the FORK system call to create a new process called a child process. Parent process sorts the integers using a sorting algorithm and waits for the child process using WAIT system call to sort the integers using any sorting algorithm. Also demonstrate zombie and orphan states	4
3	Implement the C program to simulate any 2 CPU Scheduling Algorithms (1 preemptive and 1 non- preemptive) with different arrival times.	2
4	Implement the C program for Producer Consumer problem using counting semaphores and mutex/binary semaphore	2
5	Implement the C program for Reader-Writer problem with reader priority.	2
6	Implement the C program for Deadlock Avoidance Algorithm: Bankers Algorithm.	2
7	Implement the C program for the Dining Philosophers problem.	2
8	Implement the C program for any one Page Replacement Algorithm with minimum three frames as an input	4
9	Implement the C program for any one Disk Scheduling Algorithm	2
10	Implement Full duplex communication between two independent processes. First process accepts sentences and writes on the first pipe to be read by the second process. Second process counts the number of characters, number of words and number of lines in accepted sentences, and writes the contents on the second pipe to be read by the first process and displays on standard output.	4
11	Implement a new system call in the kernel space, add this new system call in the Linux kernel by the compilation of this kernel (any kernel source, any architecture and any Linux kernel distribution) and demonstrate the use of this embedded system call using C program in user space.	4
Total Hours		30
Text Books		
1. William Stallings, Operating System: Internals and Design Principles, Prentice Hall, 8th Edition, 2014, ISBN-10: 0133805913 • ISBN-13: 9780133805918 2. Abraham Silberschatz, Peter Baer Galvin and Greg Gagne, Operating System Concepts, John Wiley & Sons, Inc., 9th Edition, 2012, ISBN 978-1-118-06333-0 3. Arnold Robbins, Nelson H. F. Beebe, Classic Shell Scripting, O'Reilly Media, Inc., 2005, ISBN 9780596005955		
Reference Book		
1. Tom Adelstein and Bill Lubanovic, Linux System Administration, O'Reilly Media, ISBN-10: 0596009526, ISBN-13: 978-0596009526. 2. Harvey M. Deitel, Operating Systems, Prentice Hall, ISBN-10: 0131828274, ISBN-13: 978- 0131828278. 3. Thomas W. Doeppner, Operating System in depth: Design and Programming, WILEY, ISBN: 978- 0-471-68723-8. 4. Mendel Cooper, Advanced Shell Scripting, Linux Documentation Project. 5. Andrew S. Tanenbaum & Herbert Bos, Modern Operating System, Pearson, ISBN-13: 9780133592221, 4th Edition.		
Online References		
1. https://onlinecourses.nptel.ac.in/noc25_cs78/preview 2. https://www.e-booksdirectory.com/details.php?ebook=9907 3. https://www.scaler.com/topics/course/free-operating-system-course/		

Second Year B. Tech Information Technology															
Spring Semester															
Course Code: B25IT403								Course Name: Computer Network							
Teaching Scheme (Hours/Week)					Examination Scheme						Credits				
L	P	T	OL	ODL	CIE	ETE	TW	PR	OR	TOTAL	L	P	T	TOTAL	
3	-	-	-	-	40	60	-	-	-	100	3	-	-	3	
Course Outcomes: After completing the course, the students will be able to: CO1: Explore concepts and fundamentals of data communication and computer networks. CO2: Design the network using IP addressing and subnetting / supernetting schemes CO3: Implement client server Program using Sockets CO4: Compare protocols at application layer CO5: Design wireless network															
Unit	Contents														Hrs.
1	Basics to Computer Network: Introduction to Computer Network and TCP/IP Protocol suite. Data Link Layer: Data Link Layer Design issues and Services, Framing: fixed-size framing, variable size framing. Random access: ALOHA, CSMA, CSMA/CD and CSMA/CA Error Detection and Correction, Linear Block Codes: hamming code, Hamming Distance, parity check code. Cyclic Codes: CRC (Polynomials), Advantages of Cyclic Codes, Other Cyclic Codes (Examples: CHECKSUM: One's Complement, Internet Checksum). Flow control protocols. Noiseless channels: simplest protocol, stop-and-wait protocol, Noisy channels: stop-and-wait Automatic Repeat Request (ARQ), go-back-n ARQ, Selective repeat ARQ, piggybacking.														8
2	Network Layer: Services, IPv4 Addresses: Classful and Classless Addressing, Special Addresses, NAT, Subnetting, Supernetting, Delivery and Forwarding of IP Packet, Structure of Router, IPv4: Fragmentation, Options, Checksum, ARP: Address Mapping, ARP Protocol, RARP, ICMPv4, Unicast Distance Vector Routing, Link State Routing, Unicast Routing Protocols: RIP, EIGRP, OSPF, BGP, IPv6 Addressing.														8
3	The Transport Protocols: UDP: Datagram, Services, Applications, TCP: Transport service primitives, Features, Segment, TCP Connection management Handshake, TCP state transition, TCP Timers, Flow control, Congestion Control Algorithms, Leaky Bucket, Sockets: Socket primitives, TCP timers, TCP Flow control (sliding Window), TCP Congestion Control: Slow Start														8
4	Application Layer Protocols: DHCP, DNS Name Space, Resource Record and Types of Name Server., FTP, TFTP, HTTP, SMTP, POP, IMAP, MIME, Telnet Network Management: SNMP, Telnet.														8
5	Wireless Networks: IEEE 802.3 Infrastructure Network and Infrastructure-less, MANET: Issues in Adhoc Wireless Network, Adhoc Network MAC Layer: Design Issues, Design Goal, Classification, MACAW Routing: Issues in Designing a Routing Protocol for Ad-hoc Wireless Networks - Classifications of Routing Protocols, DSDV, AODV, DSR, Wireless Sensor Network: Architecture, Design Issues and Challenges Sensor network protocols														7
Total Hours														39	
List of Activities															
1. Your company has been assigned the network 192.168.10.0/24. It consists of five departments, for each subnet, specify Network address, Subnet mask, First usable IP, Last usable IP, Broadcast address ○ HR: Requires 25 devices. ○ Finance: Requires 50 devices.															

- IT: Requires 60 devices.
 - Sales: Requires 30 devices.
 - R&D: Requires 40 devices
2. You are a network engineer tasked with optimizing the routing configuration of a distributed organization. The organization owns several IP networks that need to be aggregated into a single supernet for simplified routing. Specify the supernet's network address, the subnet mask in dotted decimal notation. The networks to be combined are:
 - 192.168.0.0/24
 - 192.168.1.0/24
 - 192.168.2.0/24
 - 192.168.3.0/24
 3. Conduct an in-depth on-site assessment of the campus network infrastructure to gain first hand insights into its configuration
 4. Introduction to server administration and configuration of Web Server.
 5. Study of WLAN with static IP addressing and DHCP with MAC security and filters.

Text Books

1. A.S. Tanenbaum, Computer Networks, 4th edition Pearson Education
2. B. A. Forouzan, Data Communications and Networking, 5th edition, TMH
3. C. Siva Ram Murthy, B. S. Manoj, Adhoc Wireless Networks: Architecture and Protocols, Pearson Education, ISBN: 978-81-317-0688-6, 1st Edition.

Reference Book

1. S.Keshav, An Engineering Approach To Computer Networking, Pearson
2. Behrouz A. Forouzan, TCP/IP Protocol Suite, McGraw Hill Education, ISBN: 978-0-07-070652-1, 4th Edition.

Online References

1. <https://www.netacad.com/courses/networking/networking-essentials>
2. <https://www.coursera.org/learn/computer-networking>
3. <https://nptel.ac.in/courses/106105081>

Second Year B. Tech Information Technology															
Spring Semester															
Course Code : B25IT404								Course Name: Foundation of Web Technology							
Teaching Scheme (Hours/Week)					Examination Scheme						Credits				
L	P	T	OL	ODL	CIE	ETE	TW	PR	OR	TOTAL	L	P	T	TOTAL	
3	-	-	-	-	40	60	-	-	-	100	3	-	-	3	
Course Outcomes : After completing the course, the students will be able to: CO1: Enumerate the fundamental concepts of web programming. CO2: Develop Static and Dynamic websites using technologies like HTML, CSS, Bootstrap. CO3: Demonstrate the use of web scripting languages. CO4: Design and develop simple web pages using a combination of HTML, CSS, and Bootstrap CO5: Evaluate the usability and accessibility of web pages developed using jQuery															
Unit	Contents														Hrs.
1	Introduction to Web Technologies Introduction to the World Wide Web: History and evolution of the internet and the World Wide Web, Basic concepts: URLs, HTTP, browsers, Client-server architecture Introduction to HTML : Basic structure of an HTML document DOCTYPE, head, body ,Common HTML tags: headings, paragraphs, lists, images, links, Attributes and their usage, Semantic HTML5 elements e.g., header, footer, nav, article ,Creating tables and forms Introduction to Multimedia: Embedding audio and video in HTML, Image optimization techniques Introduction to the W3C : Role of the W3C in web standards development, W3C recommendations and specifications, Importance of following web standards														7
2	Cascading Style Sheets CSS Introduction to CSS : Basic syntax and structure of CSS,Inline, internal, and external stylesheets,Selectors element, class, ID CSS Properties: Text formatting font, color, size , Backgrounds color, images , Borders, Margins and padding, Display and visibility, Positioning static, relative, absolute, fixed CSS Layout Techniques: Flexbox-Understanding and implementing flexbox layouts, Grid Layout- Creating complex grid-based designs, Responsive design principles and techniques CSS Animations and Transitions: Creating simple animations using CSS, Applying transitions to CSS properties														7
3	Bootstrap Framework Introduction to Bootstrap : Overview of Bootstrap and its features, Installing and using Bootstrap, Bootstrap grid system Bootstrap Components: Using Bootstrap components: buttons, forms, tables, navigation bars, etc., Customization of Bootstrap components Responsive Design with Bootstrap: Creating responsive layouts using Bootstrap classes, Media queries and responsive breakpoints Bootstrap JavaScript Plugins: Using Bootstrap JavaScript plugins e.g., modals, carousels, tooltips														7
4	Web Scripting Languages Introduction to Scripting Languages: Scripting languages, Comparison of JavaScript with other scripting languages Introduction to JavaScript : Basic syntax and structure of JavaScript, Variables, data types, operators, Control flow statements if-else, loops , Functions and function scope Working with JavaScript : Arrays and objects in JavaScript, DOM manipulation using JavaScript Event handling click, mouseover, etc. , Advanced JavaScript Concepts JSON: Introduction to JSON, parsing and stringifying JSON data, Asynchronous JavaScript: Promises, async/await, Error handling in JavaScript Introduction to jQuery : JQuery, Basic jQuery syntax and selectors,DOM manipulation with jQuery,jQuery effects and animations,Using jQuery plugins														12

5	AJAX and Web Services Introduction to AJAX : Concept of AJAX , XMLHttpRequest object ,Making HTTP requests with AJAX, Data Exchange with AJAX, Sending and receiving data using AJAX, Handling AJAX responses, JSON as data exchange format AJAX Error Handling: Handling AJAX errors and network issues Introduction to Web Services: RESTful APIs and their use, Consuming web services with JavaScript and jQuery	6
Total Hours		39
List of Activities		
1. Create a simple HTML page including: Headings, Paragraphs, Images, Links, Table, Form. 2. Design and implement a simple CSS animation. 3. Explore and use Bootstrap JavaScript plugins e.g., modals, carousels . 4. Create a dynamic web page using jQuery that responds to user interactions. 5. Research and explore different types of web services RESTful APIs, SOAP . 6. Consume a web service using JavaScript or jQuery.		
Text Books		
1. Kogent Learning Solutions Inc, Web Technologies: HTML, JAVASCRIPT, PHP, JAVA, JSP, XML and AJAX, Blackbook, Dreamtech Press, Second Edition, ISBN: 9788177228496. 2. 2. Raymond Camden, Andy Matthews, JQuery Mobile Web Development Essentials, Packt Publishing, Second Edition, 9781782167891. 3. Steven M. Schafer, "HTML, XHTML and CSS", Wiley India Edition, Fourth Edition, 978- 81- 265-1635-3		
Reference Book		
1. Dr.HirenJoshi, Web Technology and Application Development, DreamTech, First,ISBN:978-93- 5004-088-1 2. Steven M. Schafer, "HTML, XHTML and CSS", Wiley India Edition, Fourth Edition, 978- 81-265- 1635-3 3. Ivan Bayross,"Web Enabled Commercial Application Development Using HTML, JavaScript, DHTML and PHP,BPB Publications,4th Edition,ISBN:978-8183330084. 4. Brain Fling, Mobile Design and Development, O'REILLY, First Edition, ISBN: 13:978-81- 8404-817-9 5. Adam Bretz& Colin J Ihrig, Full Stack Javascript Development with MEAN, SPD, First Edition, ISBN:978-0992461256. 6. JavaScript: The Definitive Guide - Master The World's Most-Used Programming Language, Seventh Edition 7. Java Script, D.Flanagan, O'Reilly, SPD. 8. Programming Typescript: Making YourJavaScript Applications Scale, Boris Cherny		
Online References		
1. https://nptel.ac.in/courses/106105084 2. https://onlinecourses.swayam2.ac.in/nou24_cs09/preview 3. https://www.coursera.org/specializations/html-css-javascript-for-web-developers 4. https://www.coursera.org/specializations/web-design 5. Learning Amazon Web Services AWS - A Hands-on Guide to the Fundamentals of AWS Cloud Author: Mark Wilkins. 2. https://www.meanacademy.in/web-technologies 3.		

Second Year B.Tech Information Technology														
Spring Semester														
Course Code : B25IT405					Course Name: Website Development and Hosting									
Teaching Scheme (Hours/Week)					Examination Scheme						Credits			
L	P	T	OL	ODL	CIE	ETE	TW	OR	PR	TOTAL	L	P	T	TOTAL
-	4	-	-	-	-	-	50	-	-	50	-	2	-	2
Course Outcomes : After completing the course, the students will be able to: CO1:Understand the fundamental concepts in designing web based applications and apply frontend and backend technologies CO2: Apply the client side and server side technologies for web application development CO3:Analyze the web technology languages, frameworks and services CO4:Create the effective web applications for business functionalities using latest web development platforms														
Contents in Association with Elite Solutions														
Content related to assignments: Website planning, Front End Development (HTML, CSS, Bootstrap, Validation and Design Layout Development.),Responsive Website Development, Introduction of Bootstrap, Bootstrap Components and applying on web pages, JAVA Script, Connectivity using JDBC driver, Website hosting terminologies														
Sr. No	Activities										Duration			
1	Website planning: Students should visit different websites (Min. 7) for the different client projects and note down the evaluation results for these websites. Refer the following points for the same: Website URL,Purpose of Website, Things liked in the website, Things disliked in the website,Look and Feel, Target audience,Navigation, Content Representation,Responsiveness, Overall evaluation of the website (Remarks)										4			
2	Design an attractive web page for an online food delivery system using HTML and CSS, craft a user-friendly interface with navigation, engaging content sections, and responsive design to enhance the overall dining experience. Also add the basic pages "Home," "About," "Menu," and "Contact Us," featuring the company logo "Online FoodShop".										4			
3	Design and Build a Survey/Feedback Form for food delivery systems using HTML and CSS. The form should include various questions for users to answer. Use HTML to design the basic layout of the form, and CSS to enhance the design with elements such as text decoration, text color, background color, text alignment, margin, padding, and more.										4			
4	A)Implement an online food delivery application in Java Script using following: a) Design UI of application using HTML, CSS etc. b) Include Java script validation c) Use of prompt and alert window using Java Script B) Design a Student Grade Calculator using HTML CSS and JavaScript.										4			
5	Design a User login page for online food ordering system with entries for name, mobile number email id and login button and perform following validations. a. Validation for correct names b. Validation for mobile numbers c. Validation for email id d. Validation if no entered any value e. Congratulations and welcome page upon successful entries										4			
6	Write a JDBC Program in JAVA to perform following operations on the online food ordering system database. Add record Delete record Update record Display record										4			
7	Create a landing page for a website of your choice using bootstrap.										4			
8	Study and Write the following terminologies for hosting a website: DNS, ICANN, domain ownership, name server, epp code, domain transfer process, domain name buying and selling process, What is hosting, types of hosting, difference										4			

	between shared, VPS, dedicated and Cloud hosting, cpanel, starting own domain name and hosting service without any investment.	
Total Hours		32
Project Based Learning (PBL) Topics		
P1	Develop a web app that allows users to track their income, expenses, and budget. Users can categorize transactions, set financial goals, and visualize financial health.	
P2	Build a blog platform where you can write and publish articles. Showcase your thoughts, experiences, and expertise. Use HTML, CSS, JavaScript (for dynamic features)	
P3	Build a travel booking platform where users can search for flights, hotels, and tours. Implement features like personalized recommendations, booking management, and payment gateways. Use JDBC connectivity for the same.	
P4	Develop a professional portfolio website for showcasing your skills, projects, and resume. Include sections like an about me page, project showcase, skills, and contact information. Use HTML for content structure, CSS for styling, and consider adding smooth scrolling animations.	
P5	Create an end-to-end event management system for organizing conferences, workshops, or festivals. Allow event organizers to create events, manage registrations, and track attendance. Implement features like ticketing, notifications, and analytics.	
P6	Develop a fun quiz app with 10 randomly generated questions from different categories to easily play and test your knowledge.	
P7	Develop a Weather Forecasting Website. To help people check weather conditions, developers can create a weather forecast website with this web dev project. They can develop different features to check weather conditions, humidity, current temperature, and wind speed.	
P8	Develop a web-based dashboard that interacts with various smart home devices. Users should be able to view the status of each device (on/off, temperature, security alerts) and control them directly from the dashboard.	
Total Hours		28
Text Books		
1. Jeffrey C. Jackson, "Web Technologies: A Computer Science Perspective", Second Edition, Pearson Education, 2007, ISBN 978-0131856035 2. DT Editorial Services "HTML 5 Black Book (Covers CSS3, JavaScript, XML, XHTML, AJAX, PHP, jQuery)" 2Ed, Dreamtech Press. 3. Laura Lemay, Rafe Colburn and Jennifer Kyrnin, "Mastering HTML, CSS & Javascript Web Publishing", SAMS, BPB Publications		
Reference Books		
1. Marty Hall, Larry Brown, "Core Web Programming", Second Edition, Pearson Education, 2001, ISBN 978-0130897930. 2. <u>Kathy Sierra</u>, <u>Bert Bates</u>, <u>Trisha Gee</u>, "Head First JAVA", Third Edition, O'Reilly Media, Inc, ISBN: 9781491910771 3. The Web Application Hacker's Handbook: Finding and Exploiting Security Flaws 2nd Edition, Kindle Edition.		
Online References		
1. HTML, The Complete Reference http://www.htmlref.com/ 2. https://getbootstrap.com/		

Second Year B. Tech Information Technology														
Spring Semester														
Course Code: B25IT406					Course Name: Environmental Sustainability									
Teaching Scheme (Hours/Week)					Examination Scheme						Credits			
L	P	T	OL	ODL	CIE	ETE	TW	PR	OR	TOTAL	L	P	T	TOTAL
-	-	2	-	-	-	-	50	-	-	50	-	-	2	2
Course Outcomes: On completion of the course, learners will be able to: CO1: Explain sustainability concepts, goals, and associated challenges. CO2: Analyse pollution sources, impacts, and sustainable waste management strategies. CO3: Apply carbon management and sustainable habitat principles. CO4: Evaluate renewable energy sources for sustainable solutions. CO5: Assess environmental issues and conservation strategies through case studies.														
Unit		Contents												Hrs.
1		Introduction: Need and Concept of Sustainability, Social, Environmental, and Economic Sustainability Concepts, Sustainable Development, Overview of the 17 Sustainable Development Goals (SDGs),Nexus between Technology and Sustainable Development, Challenges for Sustainable Development., Multilateral Environmental Agreements and Protocols, Clean Development Mechanism (CDM),Environmental Legislations in India (Water Act, Air Act).												6
2		Pollution: Definition, types, Air Pollution- Sources and Effects, Water Pollution- Sources, Sustainable Wastewater Treatment, Solid Waste, Sources and Impacts of Solid Waste, Zero Waste Concepts (3R Concept), Global Environmental Issues, Resource Degradation, Climate Change, Global Warming, Ozone Layer Depletion, Regional and Local Environmental Issues. Shares												6
3		Carbon Management: Carbon Credits and Carbon Trading, Carbon Footprint. Sustainable Habitat, Basic Concepts of Sustainable Habitat, Green Buildings and Green Materials for Building Construction, Material Selection for Sustainable Design, Green Building Certification, Methods for Increasing Energy Efficiency of Buildings, Sustainable Cities and Sustainable Transport												6
4		Renewable Energy and Sustainable Solutions: Energy Sources, Basic Concepts of Conventional and Non-Conventional Energy, Solar Energy, Fuel Cells, Wind Energy, Small Hydro Plants, Bio-Fuels, and Energy Derived from Oceans, Geothermal Energy.												7
5		Role of Individuals, Corporate, and Society: Environmental Values, Positive and Adverse Impact of Technological Developments on Society and Environment, Role of an Individual/ Corporate/ Society in Environmental Conservation. Case Studies: The Bhopal Gas Tragedy, New Delhi’s Air Pollution, Arsenic Pollution in Groundwater (West Bengal),Narmada Valley Project, Cauvery Water Dispute,Fukushima Daiichi Disaster (Japan),Ozone Hole over Antarctica,Ganga Pollution, Deterioration of Taj Mahal,Uttarakhand Flash Floods												7
Total Hours													32	
List of Activities														
1. Research and present a case study on a successful implementation of one of the 17 Sustainable Development Goals (SDGs) in a specific region. 2. Design a plan for sustainable wastewater treatment for a small community or institution, considering local conditions and available resources. 3. Develop a presentation on the advantages and disadvantages of different renewable energy sources, considering factors like technology, cost, environmental impact, and scalability. 4. Research and present a case study on a successful green building project, highlighting its sustainable features and environmental benefits. 5. Organize a waste audit within your school/community to assess the types and quantities of waste														

generated. Develop a waste management plan based on the 3R concept (Reduce, Reuse, Recycle).

Text Books

1. Rogers, P.P., Jalal, K.F., Boyd, I.A. (Latest Edition) - An Introduction to Sustainable Development. Earthscan
2. Kalam, A.P.J. (Latest Edition) - Target 3 Billion: Innovative Solutions Towards Sustainable Development. Penguin Books
3. Kaushik, A. and Kaushik (Latest Edition) - Perspectives in Environmental Studies. New Delhi: New Age International Publishers
4. Dhameja, S.K. (Latest Edition) - Environmental Studies. S.K. Kataria and Sons, New Delhi
5. Bharucha, E. (Latest Edition) - Environmental Studies for Undergraduate Courses. New Delhi: University Grants Commission
6. Wright, R.T. (Latest Edition) - Environmental Science: Towards a Sustainable Future. New Delhi: PHL Learning Private Ltd.
7. Rajagopalan, R. (Latest Edition) - Environmental Studies. New York: Oxford University Press

Reference Books

1. R. Rajagopalan (2006) - Environmental Studies. Oxford University Press
2. M. AnjiReddy (2006) - Textbook of Environmental Sciences and Technology. BS Publication
3. Richard T. Wright (2008) - Environmental Science: Towards a Sustainable Future. PHL Learning Private Ltd., New Delhi
4. Gilbert M. Masters and Wendell P. Ela (2008) - Environmental Engineering and Science. PHL Learning Pvt. Ltd.
5. Daniel B. Botkin and Edward A. Keller (2008) - Environmental Science. Wiley India Edition
6. Anubha Kaushik (2009) - Environmental Studies. New Age International Publishers

Second Year B. Tech Information Technology														
Spring Semester														
Course Code : B25IT407					Course Name: Startup Fundamentals and Financing									
Teaching Scheme (Hours/Week)					Examination Scheme						Credits			
L	P	T	OL	ODL	CIE	ETE	TW	PR	OR	TOTAL	L	P	T	TOTAL
-	-	2	-	-	-	-	50	-		50	-	-	2	2
Course Outcomes:														
After completing the course, the students will be able to:														
CO1: Analyze the key concepts and frameworks shaping the innovation and start-up ecosystem.														
CO2: Develop expertise in building a startup ecosystem, understanding its key components, managing dynamics, and boosting its overall productivity.														
CO3: Evaluate the contributions of various stakeholders within the ecosystem in promoting and sustaining the growth of start-ups.														
CO4: Demonstrate an understanding of global trends in the startup ecosystem and product development.														
CO5: Analyze and compare various startup ecosystems, and develop performance indicators to evaluate their effectiveness.														
Unit	Contents													Hrs.
1	Start-up Opportunity: Identify business opportunity with problem identification, market size, existing pains for customers, existing alternatives, customer psychology, willingness to pay, customer segments.													5
2	Product/ Service Proposal: Value Proposition Canvas, problem-solution fit, brainstorming, competition analysis, creating competitive advantage, sustainable differentiation.													5
3	Business model: Types, Lean canvas, Risky assumptions related to product, market, business, and execution capabilities													5
4	Minimum Viable Product (MVP): Create and iterate, testing of MVP, customer feedback, validate risky assumptions, solution-market fit													5
5	Financial Plan and Marketing strategy: Manpower, Sales, Expenses, profitability projections, reality check, Funding plan, Pitch deck, Importance of brand and branding strategy, positioning, market penetration strategy/ plan, digital marketing, use of social media, customer acquisition Use of technology: for business scalability, effective execution, growth plan													6
													Total Hours	26
Text Books														
1. Startup and Entrepreneurship - Learn Before Start, Prasanjeet Anand,White Falcon Publishing; 1 edition (2023) 2. Startup Finance 360° - Founder's Guide to Startup Finance,ZebraLearn Pvt Ltd,ISBN-10-8196373597 3. Entrepreneurship in Action, Coulter, PHI 2nd Edition 4. Entrepreneurship Development,E. Gordon & K. Natarajan, Himalaya 5. Poornima M. Charantimath, Entrepreneurship Development and Small Business Enterprises, 3rd Edition, Pearson Education, 2018														
Reference Book														
1. The Startup Owner’s Manual: The Step-by-Step Guide for Building a Great Company,Steve Blank and Bob Dorf,K & S Ranch ISBN – 978-0984999392 2. The Lean Startup: How Today’s Entrepreneurs Use Continuous Innovation to Create Radically Successful Businesses,Eric Ries,Penguin UK ISBN – 978-0670921607 3. Entrepreneurship Development and Small Business Enterprise,Poornima Charantimath,Pearson Education; ISBN: 978-8131759196 4. Peter Thiel & Blake Masters, Zero to One: Notes on Startups, or How to Build the Future, Random House, 2014. 5. The Startup Owner’s Manual: The Step-by-Step Guide for Building a Great Company,Steve Blank and Bob Dorf,K & S Ranch ISBN – 978-0984999392														

Online References
1. https://www.entrepreneur.com/article/251687 2. https://www.forbes.com/sites/forbesbusinessdevelopmentcouncil/2020/09/14/13-key-steps-to-developing-a-go-to-market-strategy/?sh=53023c476fc1 3. https://skillindia.gov.in/ 4. https://startuptalky.com/list-of-government-initiatives-for-startups/